

Web Application Development

Lab 2

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Exercise 1:

Link to personal website: <https://nxtsukii.github.io/>

Exercise 2: Survey form design

Functionality requirements

- All required text input fields presents (First Name, Last Name, Email, Date of Birth)

```
<form>
  <div class="form-group mb-1">
    <label for="firstName">First Name</label>
    <input type="text" class="form-control w-50" id="firstName">
  </div>

  <div class="form-group mb-1">
    <label for="lastName">Last Name</label>
    <input type="text" class="form-control w-50" id="lastName">
  </div>

  <div class="form-group mb-1">
    <label for="email">Email</label>
    <input type="email" class="form-control w-50" id="email">
  </div>

  <div class="form-group mb-3">
    <label for="dob">Date of Birth</label>
    <input type="date" class="form-control w-50" id="dob">
  </div>
```

First Name

Last Name

Email

Date of Birth

- Radio buttons work correctly for “How did you hear about us?” section

How did you hear about us?

☐ Search engine ☐ Word of mouth ☒ Social Media ☐ Other

```
<h5>How did you hear about us?</h5>
<div class="form-check form-check-inline">
  <input class="form-check-input" type="radio" name="source" id="search" value="search">
  <label class="form-check-label" for="search">Search engine</label>
</div>
<div class="form-check form-check-inline">
  <input class="form-check-input" type="radio" name="source" id="word" value="word">
  <label class="form-check-label" for="word">Word of mouth</label>
</div>
<div class="form-check form-check-inline">
  <input class="form-check-input" type="radio" name="source" id="social" value="social">
  <label class="form-check-label" for="social">Social Media</label>
</div>
<div class="form-check form-check-inline mb-3">
  <input class="form-check-input" type="radio" name="source" id="other" value="other">
  <label class="form-check-label" for="other">Other</label>
</div>
```

- Checkboxes function properly for announcements preferences

Would you like to receive announcements about new CDs and special offers?

- ☒ YES, I'd like that.
- ☒ YES, please send me email announcements.

```
<h5>Would you like to receive announcements<br>about new CDs and special offers?</h5>
<div class="form-check">
  <input class="form-check-input" type="checkbox" id="like">
  <label class="form-check-label" for="like">YES, I'd like that.</label>
</div>
<div class="form-check mb-2">
  <input class="form-check-input" type="checkbox" id="emailAnnouncements">
  <label class="form-check-label" for="emailAnnouncements">YES, please send me email announce
</div>
```

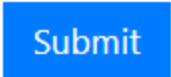
- Dropdown/select menu implemented for contact method

Please contact me by: Email or postal mail ▼

- Email or postal mail
- Email only
- Postal mail only

```
<div class="form-group">
  <label for="contactMethod">Please contact me by:</label>
  <select class="form-control" id="contactMethod">
    <option>Email or postal mail</option>
    <option>Email only</option>
    <option>Postal mail only</option>
  </select>
</div>
```

- Submit button is present



Submit

```
<button type="submit" class="btn-primary btn-submit">Submit</button>
</form>
```

Exercise 3: Implementing flexbox for a 3-column, fixed-width layout

- Defining the flex container

The class `.main` is used as the flexbox container

```
.main {
  display: flex;
  justify-content: space-between;
  border: 1px solid #e1c275;
  border-left: none;
  border-right: none;
  min-height: 250px;
  margin-bottom: 5px;
}
```

- ‘display: flex’ activates the flexbox layout model
 - ‘flex-direction: row’ arranges its child elements (two sidebars and main content) horizontally in one row
 - The border and margin ensure proper separation and spacing between sections
- Creating columns
- Inside `.main` there are three child elements

```
<div class="main">
  <div class="sidebar-a">Sidebar A</div>
  <div class="content">Main content</div>
  <div class="sidebar-b">Sidebar B</div>
</div>
```

Each of these child elements acts as a flex item

```
.sidebar-a {
  width: 150px;
  border-right: 1px solid #e1c275;
  padding: 10px;
  box-sizing: border-box;
  font-weight: bold;
}

.content {
  flex: 1;
  border-right: 1px solid #e1c275;
  padding: 10px;
  box-sizing: border-box;
  font-weight: bold;
}

.sidebar-b {
  width: 150px;
  padding: 10px;
  box-sizing: border-box;
  font-weight: bold;
}
```

- Setting fixed and flexible widths
Sidebars are assigned with fixed widths

```
.sidebar-a {  
  width: 150px;  
  border-right: 1px solid #e1c275;  
  padding: 10px;  
  box-sizing: border-box;  
  font-weight: bold;  
}
```

```
.sidebar-b {  
  width: 150px;  
  padding: 10px;  
  box-sizing: border-box;  
  font-weight: bold;  
}
```

Main content area is made flexible:

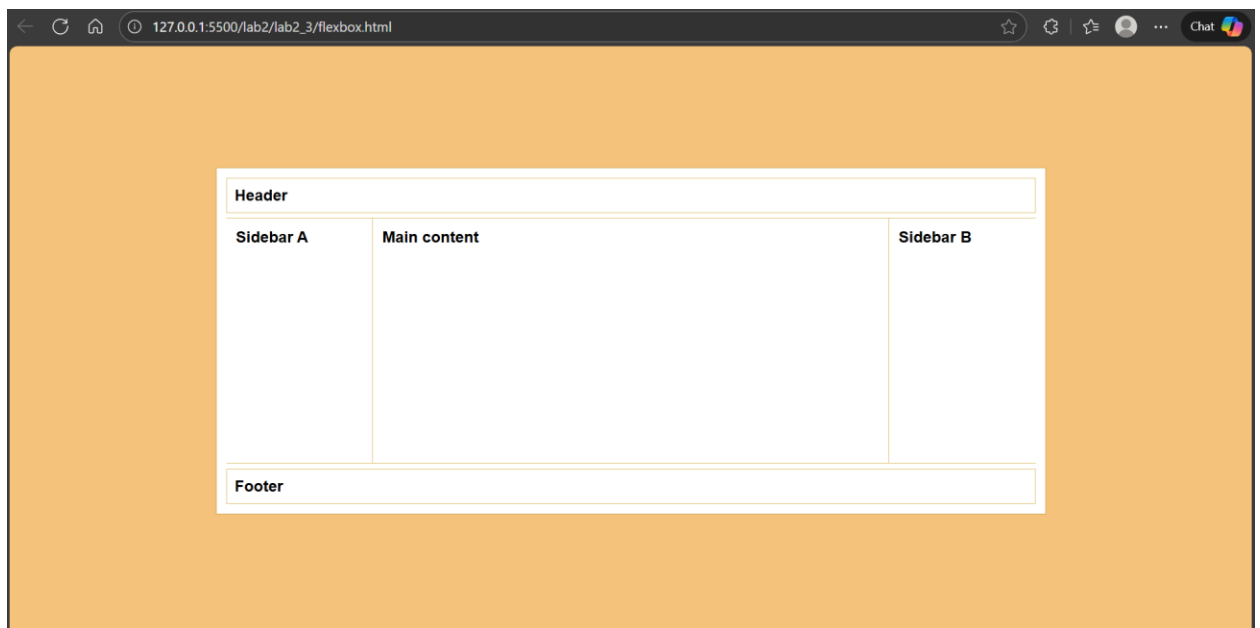
```
.content {  
  flex: 1;  
  border-right: 1px solid #e1c275;  
  padding: 10px;  
  box-sizing: border-box;  
  font-weight: bold;  
}
```

- ‘flex: 1’ tells the content area to grow and fill all remaining space after the sidebars take their fixed widths.
 - This implementation help in keeping the proportions of the layout balanced while also maintaining a fixed overall layout width.
- Fixed-width container
The webpage is centered by using a fixed-width ‘.container’

```
.container {  
  background-color: #fff;  
  width: 850px;  
  border: 1px solid #cfa34f;  
  box-sizing: border-box;  
  display: flex;  
  flex-direction: column;  
  padding: 10px;  
}
```

The flexbox layout is then used with ‘flex-direction: column’ to stack the header, main section, and footer vertically.

- Final output:



Exercise 4:

1. Flexbox implementation
 - Each column in the main section is defined as a flexbox child
- The left content takes two-thirds of the width using:

```
✓ .content {  
  flex: 2;  
  margin-right: 20px;  
}
```

The right sidebar (lecture notes) takes one-third:

```
✓ .sidebar {  
  flex: 1;  
  background-color: #f8e5b0;  
  padding: 15px;  
  margin-bottom: 40px;  
}
```

The combination of 'flex: 2' and 'flex: 1' maintains a consistent proportion between content and sidebar even if the page width changes slightly.

- Header and footer

Header uses flexbox to align the logo and text horizontally:

```
✓ .header {  
  display: flex;  
  align-items: center;  
  color: #3b240b;  
  padding: 15px;  
  border-bottom: 3px solid #cfa960;  
}
```

- 'display: flex' arranges the logo and title side-by-side (horizontally)
- A border-bottom provides a separation visually between header and body sections

The footer is placed at the bottom using standard block layout, styled to span the full width:

```

/* Footer */
.footer {
  background-color: #fff7e6;
  text-align: right;
  border-top: 1px solid #cfa960;
  padding: 10px;
  font-size: 14px;
  color: #444;
}

```

2. Final output:

